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United States Patent	12382856
Kind Code	B2
Date of Patent	August 12, 2025
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Method for locating and planting sentinel plants

Abstract

A planter configured for planting a sentinel plant in a field. The planter comprises a first planting system that is coupled to the planter. The first planting system is configured for planting a first sentinel plant having a first sentinel plant characteristic. A global positioning system is communicatively coupled to the planter. The global positioning system is configured for generating a location signal indicative of a location of the planter. A control system is communicatively coupled to the planter. The control system is configured to receive the first sentinel plant characteristic, receive the location signal, receive a georeferenced field characteristic, and control the first planting system based on at least one of the first sentinel plant characteristic, the location, or the georeferenced field characteristic.

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Appl. No.:	17/823576
Filed:	August 31, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240065138 A1	Feb. 29, 2024

Publication Classification

Int. Cl.: **A01C14/00** (20060101); **A01B69/04** (20060101); **A01C5/04** (20060101); **G05D1/00** (20240101)

U.S. Cl.:

CPC **A01C14/00** (20130101); **A01B69/008** (20130101); **A01C5/04** (20130101); **G05D1/0094** (20130101);

Field of Classification Search

CPC: A01C (14/00)

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Background/Summary**FIELD OF THE DISCLOSURE**

(1) The present disclosure generally relates to a planter and more particularly to a planter and a method for planting sentinel plants in a field.

BACKGROUND OF THE DISCLOSURE

(2) Planters are generally used to plant plants in a field by delivering a seed from a hopper to the ground with a seed delivery system. Multi-variety planters can simultaneously plant seeds from two or more different plant genotypes in a single planting operation.

SUMMARY OF THE DISCLOSURE

(3) In one embodiment, a planter configured for planting a sentinel plant in a field is disclosed. The planter comprises a first planting system that is coupled to the planter. The first planting system is configured for planting a first sentinel plant having a first sentinel plant characteristic. A global positioning system is communicatively coupled to the planter. The global positioning system is

configured for generating a location signal indicative of a location of the planter. A control system is communicatively coupled to the planter. The control system is configured to receive the first sentinel plant characteristic, receive the location signal, receive a georeferenced field characteristic, and control the first planting system based on at least one of the first sentinel plant characteristic, the location, or the georeferenced field characteristic.

(4) In another embodiment, a multi-variety planter configured for planting a sentinel plant in a field is disclosed. The multi-variety planter comprises a first planting system that is coupled to the multi-variety planter. The first planting system is configured for planting a first sentinel plant having a first sentinel plant characteristic. A second planting system is coupled to the multi-variety planter. The second planting system is configured for planting a second sentinel plant having a second sentinel plant characteristic. A global positioning system is communicatively coupled to the multi-variety planter. The global positioning system is configured for generating a location signal indicative of a location of the multi-variety planter. A control system is communicatively coupled to the multi-variety planter. The control system is configured to receive the first sentinel plant characteristic, receive the second sentinel plant characteristic, receive the location signal, receive a georeferenced field characteristic, and control at least one of the first planting system or the second planting system based on at least one of the first sentinel plant characteristic, the second sentinel plant characteristic, the location, or the georeferenced field characteristic.

(5) In yet another embodiment, a method of controlling a planter for planting a sentinel plant in a field is disclosed. The method comprises receiving a first sentinel plant characteristic. Receiving a location signal indicative of a location of the planter. Receiving a georeferenced field characteristic. Controlling a first planting system to plant the first sentinel plant based on at least one of the first sentinel plant characteristic, the location, or the georeferenced field characteristic.

(6) Other features and aspects will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a top schematic view of a tractor and a planter having a plurality of row units;

(2) FIG. 2 is a side view of one of the plurality of row units of the planter of FIG. 1;

(3) FIGS. 3-6 are schematic side views of a portion of the row unit of FIG. 2 cycling through a series of positions;

(4) FIGS. 7-10 are schematic side views of a planting assembly of the row unit of FIG. 2 cycling through a series of positions;

(5) FIGS. 11-14 are schematic side views of a portion of the row unit of FIG. 2 cycling through a series of positions under alternate conditions;

(6) FIG. 15 is a block diagram of a planter; and

(7) FIG. 16 is a flow diagram of a method for operating a planter in a field.

(8) Before any embodiments are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Further embodiments of the invention may include any combination of features from one or more dependent claims, and such features may be incorporated, collectively or separately, into any independent claim.

(9) As used herein, unless otherwise limited or modified, lists with elements that are separated by conjunctive terms (e.g., “and”) and that are also preceded by the phrase “at least one of” or “one or more of” indicate configurations or arrangements that potentially include individual elements of the

list, or any combination thereof. For example, “at least one of A, B, and/or C” or “one or more of A, B, and/or C” indicates the possibilities of only A, only B, only C, or any combination of two or more of A, B, and C (e.g., A and B; B and C; A and C; or A, B, and C).

DETAILED DESCRIPTION

(10) FIG. 1 illustrates a planter **10** for planting plants **11** (FIG. 6) in seed form including sentinel seeds **65** (FIG. 6). The planter **10** may be a high-speed planter, an in-furrow planter, a multi-variety planter, a seedling planter, an air seeder, a broadcast seeder, or other type of planter. A first planting system **12** for planting one variety of plant **11** in the form of a seed may be coupled to the planter **10** or the planter **10** may be a multi-variety planter and also have a second planting system **14** for planting more than one variety of plant **11** in the form of a seed. The planter **10** for planting seeds is disclosed in commonly assigned U.S. patent application Ser. No. 17/107,041, the entire contents of which are hereby incorporated by reference. Other implementations of the first planting system **12** or the second planting system **14** for planting plants **11** in seedling form are contemplated by this disclosure such as that disclosed in commonly assigned U.S. patent application Ser. No. 16/511,918, the entire contents of which are hereby incorporated by reference.

(11) The illustrated planter **10** is selectively coupled to a vehicle **15**, such as a tractor, to be pulled across a field **20** or ground (FIG. 2). The planter **10** includes a planter frame **25**. The illustrated first planting system **12** and second planting system **14** each include at least one planting row unit **30** and at least one hopper **35** containing seeds to be planted, which are coupled to the planter frame **25**. In some implementations the seeds are all the same while in other implementations, the seeds may be different by hopper **35** and may be planted differently by row or by location within a row. In other examples, without limitation, the planter frame **25** may be rigidly attached to a prime mover such as with a three-point hitch of vehicle **15** or planter frame **25** may be permanently integrated with vehicle **15**.

(12) With reference to FIG. 2, one of the plurality of row units **30** is shown in greater detail. The row units **30** may be identical to each other in some implementations but may have some differences in other implementations. Each row unit **30** is provided with a central frame member **40** supported on the planter frame **25** by downforce control actuators. The central frame member **40** is configured to support a seeding system **45**, a press wheel **50**, and a mini hopper **55**. The seeding system **45** is operable to create a hole **60** in the field **20** or ground for receiving a sentinel seed **65** (FIG. 7) and to inject the sentinel seed **65** into the space. The press wheel **50** then fills in the space with dirt. The seeding system **45** is configured to create a plurality of the holes **60** in series, the holes **60** being discrete from each other. The discrete holes **60** are in contrast with other planters configured to create trenches. The seeding system **45** includes a planting assembly **70**, a driving assembly **75**, and a support assembly **77**.

(13) In the illustrated implementation, the row unit **30** includes a single seeding system **45**. However, in some cases it may be desirable to include a second seeding system **45** on some or all of the row units **30**. The second seeding system **45** would be mounted to the central frame member **40** in line with the first seeding system **45** (e.g., such that both seeding systems **45** are configured to plant a single row). The second seeding system **45** may include its own driving assembly **70** or may be connected to all or part of the driving assembly **70** of the first seeding system **45**. In yet other implementations, each row unit **30** may include three or more seeding systems **45**.

(14) As best illustrated in FIGS. 7-10, the planting assembly **70** includes a seed guide **80** and a spike **85**. The seed guide **80** is formed as a hollow tube **90** having a guide first end **95** and guide second end **100** (FIG. 3). The guide first end **95** is selectively disposed adjacent the ground **20** and the guide second end **100** is connected to the driving assembly **75**. The hollow tube **90** includes a tube wall **105** having a seed aperture **110** disposed between the guide first end **95** and the guide second end **100**. The seed aperture **110** may include a through-hole **115** disposed in the tube wall **105**. The seed guide **80** includes a guide channel **120** (FIG. 10) extending through the hollow tube **90**. The guide channel **120** may be formed as a through-hole defined by the tube wall **105**. The

spike **85** is formed as a shaft **125** (such that the terms “spike” and “shaft” may be used interchangeably herein) with a sidewall **130** extending between a spike first end **135** (FIG. **9**) and a spike second end **140** (FIG. **3**). The sidewall **130** is elongated and extends parallel to reciprocating movement of the spike **85**. The spike first end **135** is tapered to a point **145** such that when inserted into the ground **20**, the first end creates the hole **60** (which may also be referred to herein as a discrete hole **60**), which may be tapered. The hole **60** may be a blind hole (such as a bore or a space or a pocket) in the ground **20** for receiving one of the sentinel seeds **65**. The spike second end **140** is coupled to the driving assembly **75**, as will be described in greater detail below. The spike **85** is slidably disposed within seed guide **80**, such that the guide channel **120** receives the shaft **125**. The spike **85** and seed guide **80** are both moveable independently with respect to the central frame member **40**. The spike **85** is movable along a stroke path with respect to the seed guide **80**. The spike **85** and the seed guide **80** are coaxial. In a first position (FIG. **7**) along the stroke path, the spike first end **135** extends lower than the guide first end **95**. In the first position, the seed aperture **110** is blocked by the spike **85** such that the seed aperture **110** is not in fluid communication with the guide channel **120** of the seed guide **80**. In a second position (FIG. **9**) along the stroke path, the spike first end **135** is disposed above the seed aperture **110** of the seed guide **80**. In the second position, the seed aperture **110** is open (not blocked by the spike **85**) such that the seed aperture **110** is in fluid communication with the guide channel **120** of the seed guide **80** (and ultimately in fluid communication with the hole **60** in the ground **20**). The spike **85** is movable through the stroke path with respect to the seed guide **80** from the first position to the second position and back to the first position, and so on, in a reciprocating fashion as illustrated in FIGS. **7-10**. The stroke path may also include, in addition to the reciprocating component towards and away from the ground **20**, a rocking or tilting component, which will be described in greater detail below. (The terms “tilting” and “rocking” may be used interchangeably herein.) However, in some implementations, the stroke path may include only the reciprocating component.

(15) A seed tube **150** and the seed guide **80** define the flow path that transports the sentinel seed **65** to the hole **60**. The seeding system **45** is configured to retain the sentinel seed **65** outside of the stroke path of the spike **85** until the spike **85** retracts a predetermined distance. The predetermined distance may be met when the spike **85** retracts past the seed aperture **110**. In some implementations, the predetermined distance may be met by other distances when the spike **85** retracts to other points along the stroke path.

(16) With reference to FIG. **2**, the driving assembly **75** includes a motor **155**, a first crankshaft **160**, and a second crankshaft **165**. The second crankshaft **165** is operably connected to the first crankshaft **160** by a timing belt **170**. The motor **155** may be directly coupled to the first crankshaft **160** or the second crankshaft **165** with the other driven by the timing belt **170**. In some implementations, the first crankshaft **160** and second crankshaft **165** may extend the length of a single row unit **30** and drive one planting assembly **70**. In other implementations, the first crankshaft **160** and second crankshaft **165** may extend across multiple row units **30** and drive multiple planting assemblies **70**. The first crankshaft **160** is coupled to a first radial link **175** having a first length $R1$. In the illustrated implementation, the first radial link **175** may include an arm mounted on the first crankshaft **160**. In other implementations, the first radial link **175** may include a post (not shown) mounted at a radius of $R1$ to a disc (not shown) mounted on the first crankshaft **160**. Other linkages are also contemplated. The first radial link **175** is pivotally coupled to a first rigid link **180**. The first rigid link **180** may be formed in a shape designed to avoid interference with other components of the row unit **30**, as illustrated, or may be formed as a straight bar in other implementations. The first rigid link **180** is rotatably coupled to the spike second end **140** of the planting assembly **70**. Similarly, the second crankshaft **165** is coupled to a second radial link **185** having a second length $R2$ which is pivotally coupled to a second rigid link **190**. The second rigid link **190** is coupled to the guide second end **100** of the seed guide **80** of the planting assembly **70** such that the seed guide **80** is aligned coaxially with the second rigid link **190**. In some

implementations the seed guide **80** may be formed as a part of the second rigid link **190**. In other implementations, the driving assembly **75** may have other configurations for driving the spike **85** and the seed guide **80**, such as other arrangements of linkages coupled to a rotating shaft or shafts, or linear actuators (such as electric, pneumatic, or hydraulic linear actuators) may be employed. Other suitable mechanisms for driving the spike **85** and seed guide **80** are also contemplated.

(17) The support assembly **77** supports the planting assembly **70** adjacent the ground **20**. The planting assembly **70** is supported by the support assembly **77** such that the stroke path of the spike **85** is oriented such that the spike **85** reciprocates extending towards and retracting away from the ground **20** and the sidewall **130** extends parallel to the reciprocation direction of the spike **85**. The support assembly **77** includes a flexible arm **195** extending generally vertically below the central frame member **40** and a support arm **200** coupled to a lower end of the flexible arm **195** and extending generally horizontally towards the planting assembly **70**. (The term “generally” is used herein to encompass the ability to pivot, move, and/or flex from the main orientation as described below and to indicate that the main orientation need not be exact, just approximate.) The support arm **200** is coupled to the seed guide **80** of the planting assembly **70** and provides support to the planting assembly **70** and driving assembly **75** to inhibit undesired horizontal motion. The support arm **200** supports the seed guide **80** such that the seed guide **80** may reciprocate independently from the spike **85**. The horizontal support arm **200** may be spring biased toward an intermediate position but may be able to extend or retract to allow the planting assembly **70** to deflect horizontally. The flexible arm **195** is formed from a material, such as a metal (such as steel, aluminum, or other metal), polymer, or composite (such as fiberglass), that is elastic enough to bend when a force is applied and resilient enough to return to the original form when the force is released. In other implementations, the support assembly **77** may include a single rigid bar coupling the seed guide **80** to the center frame member **40**. In other implementations, the seed guide **80** is directly coupled to the central frame member **40** via a pin joint or other pivoting connection. In other implementations, the support assembly **77** may include a rigid vertical support link extending from the central frame member **40** and a horizontal support arm extending between the support link and the seed guide **80**. The support assembly **77** may have other suitable configurations in other implementations.

(18) The press wheel **50** includes an outer surface **205** and is configured to apply a downward force to the ground **20**. The press wheel **50** is supported on the central frame member **40** by a depth control actuator **210**. The depth control actuator **210** is actuatable to move the press wheel **50** vertically lower with respect to the central frame member **40** when the depth control actuator **210** is extended or vertically higher with respect to the central frame member **40** when the depth control actuator **210** is retracted. In the illustrated implementation, the press wheel **50** is supported on a linkage **215** that is rotated counterclockwise when the depth control actuator **210** is extended and clockwise when the depth control actuator **210** is retracted. Thus, the press wheel **50** has at least a component of vertical movement in response to actuation of the depth control actuator **210**. The rotation of the linkage **215** causes the vertical displacement of the press wheel **50** with respect to the central frame member **40**. The depth control actuator **210** may include a gas spring (not shown) controlled by gage wheels (not shown). In other implementations, the depth control actuator **210** may be pneumatically or hydraulically controlled. The press wheel **50** is rotated in response to a friction force between the ground **20** and the outer surface **205** when the planter **10** is moving. In other implementations, the press wheel **50** may be mounted to the central frame member **40** in other ways.

(19) The central frame member **40** supports the mini hopper **55** adjacent the seeding system **45**. The mini hopper **55** is connected to the one or more hoppers **35** and receives sentinel seed **65** therefrom. The central frame member **40** also supports a seed meter **220** such as is known in the art to provide a regulated flow of seeds. The seed meter **220** receives sentinel seeds **65** from the mini hopper **55**, separates the sentinel seeds **65**, and supplies metered sentinel seeds **65** to the seed tube (which may

also be referred to herein as a seed conduit), one after another in substantially equal sequential intervals. In some implementations, the seed meter **220** may receive sentinel seeds **65** directly from one of the hoppers **35**. The seed tube **150** may be pressurized by a pressure source **230** (e.g., a pump or other source of pressurized air) and is in fluid communication with the seed aperture **110** in the seed guide **80** (and thus fluidly connected to the guide channel **120** and to the hole **60** in the ground **20**). For example, the seed guide **80** may support an end of the seed tube **150** extending at least partially into the seed aperture **110**, or the seed tube **150** and the seed guide **80** may be coupled in any other suitable manner, e.g., using a connector between the seed tube **150** and the seed guide **80** in fluid communication with the seed aperture **110**. The pressure may be constant or may be timed to match the operation of the driving assembly **75**. In other implementations, the seed tube **150** need not be pressurized, e.g., the seed tube **150** may transport the seeds by gravity. In yet other implementations, the seed tube **150** may include a continuous loop belt for transporting the seeds to the seed guide **80**. In yet other implementations, the seed tube **150** may include a mechanism to mechanically urge the seeds out of the seed tube **150**. The seed tube **150** may form a first portion of a flow path configured to transport the seeds from the seed meter **220** towards the ground **20**. The seed guide **80** may define a second portion of the flow path, as will be described in greater detail below.

(20) With reference to FIGS. 3-6, in operation, the vehicle **15** is operated to move the planter **10** across the ground **20** in a first direction **235** and at a planter speed **S1**. Each seeding system **45** may be started simultaneously or at different times. When started, the motor **155** of the row unit **30** is activated in order to actuate the driving assembly **75**. The activation of the motor **155** is separate from the movement of the planter **10**. Because of this, a seeding speed **S2** of the seeding system **45**, as measured by the number of seeds planted per unit time, is independent from the planter speed **S1**, as measured by the distance traveled per unit time. The activated motor **155** rotates the first crankshaft **160** counterclockwise (as viewed in FIGS. 3-6). As the first crankshaft **160** rotates, the timing belt **170** transfers the motion to the second crankshaft **165**, causing the second crankshaft **165** to rotate counterclockwise. The rotation of the first crankshaft **160** and second crankshaft **165** cause the driving assembly **75** to cycle through a plurality of stages. At the start of the cycle, shown in FIG. 3, the first radial link **175** is disposed pointing to the left of the first crankshaft **160** and the second radial link **185** is disposed pointing upward of the second crankshaft **165**. The second rigid link **190** extends generally vertically from the second radial link **185** to the seed guide **80**, which also is oriented generally vertically. The spike **85** is pivotally connected to the first rigid link **180** such that it follows the orientation of the seed guide **80**. The spike **85** is disposed between the first position and the second position relative to the seed guide **80**. Both the spike first end **135** and the guide first end **95** are disposed above the ground **20**. As shown in FIG. 4, the movement of the first crankshaft **160** and second crankshaft **165** rotates the first radial link **175** and second radial link **185**, respectively. The first radial link **175** is disposed downward of the first crankshaft **160**, moving the first rigid link **180** and the spike **85** down as well. The second radial link **185** is disposed left of the second crankshaft **165**, moving the second rigid link **190** and seed guide **80** down and tilting each toward the left (as illustrated in FIG. 4). The spike **85** is now in the first position relative to the seed guide **80**. The spike **85** is in the ground **20** (meaning engaged with and displacing soil to create the hole **60**) and the seed guide **80** is adjacent the ground **20**. The cycle continues to FIG. 5 where the first radial link **175** points right and the second radial link **185** points down. The seed guide **80** is oriented generally vertically, and the spike **85** is disposed between the first and second positions. The seed guide **80** is now in the ground **20**, further displacing soil to create the hole **60**. In other words, the seed guide **80** is configured to penetrate the ground **20** to create a portion of the hole **60**. The rotation continues to the stage shown in FIG. 6 and the first radial link **175** points upward and the second radial link **185** points left. The seed guide **80** and spike **85** are tilted to the right and the spike **85** is in the second position. The seed guide **80** is adjacent the ground **20**. The cycle then continues with the first radial link **175** returning to the left

of the first crankshaft **160** and the second radial link **185** returning to the top of the second crankshaft **165** as shown in FIG. 3. The resultant motion of the planting assembly **70** in response to the driving assembly **75** is that the seed guide **80** and the spike **85** reciprocate independently with respect to each other and with respect to the central frame member **40**. In other words, the seed guide **80** is configured to reciprocate asynchronously with respect to the spike **85**. In some portions of the resultant motion, the spike **85** retracts while the seed guide **80** extends (FIG. 8) and in other portions of the resultant motion, the spike **85** extends while the seed guide **80** retracts (FIG. 10).

(21) The movement of the planting assembly **70** is shown in more detail in FIGS. 7-10. As the planter **10** travels along the row in the first direction **235** and the driving assembly **75** operates the spike **85** and seed guide **80** both move toward the ground **20**, as shown in FIG. 7. As the spike **85** pierces the ground **20**, the tapered point **145** displaces the soil with minimal force. The spike **85** is tilted away from the first direction **235** as it pierces the ground **20**, meaning the spike first end **135** is further in the first direction **235** than the spike second end **140**. At this time a sentinel seed **65** from the mini hopper **55** has been supplied to the seed tube **150** by the seed meter **220** and travels down toward the planting assembly **70**. As shown in FIG. 8, once the spike **85** has reached a maximum extension, it begins to move back toward the central frame member **40**, leaving the hole **60** in the ground **20**. The maximum extension is determined by the length R1 of the first radial link **175**. A larger length R1 creates a maximum extension that extends deeper into the ground **20**. The spike **85** is still tilted away from the first direction **235** as it travels out of the ground **20**. The seed guide **80** continues to move downward toward a maximum extension and into the ground **20**, displacing the soil further, still with minimal force required. The maximum extension of the seed guide **80** is determined by the length R2 of the second radial link **185**, in a similar manner to the maximum extension of the spike **85**. In other words, the maximum extension of the seed guide **80** defines an extended position for engaging the ground **20**. At this point, the sentinel seed **65** is disposed adjacent the seed aperture **110** and is inhibited from entering the guide channel **120** by the sidewall **130** of the spike **85**, the sidewall **130** acting as a barrier between the sentinel seed **65** and the hole **60**. As shown in FIG. 9, the spike **85** retracts, e.g., follows the stroke path away from the ground **20**. As the spike **85** retracts past the seed aperture **110**, gravity and the pressure from the seed tube **150** pushes the sentinel seed **65** into the guide channel **120** of the seed guide **80**. In some implementations, gravity may push the sentinel seed **65** into the guide channel **120** without the use of pressure. The spike **85** and the seed guide **80** are configured such that the sentinel seed **65** is blown past the spike **85** upon retraction of the spike past the seed aperture **110**. The seed aperture **110** is disposed outside of the stroke path of the spike **85** as the seed aperture **110** is disposed in the tube wall **105** of the seed guide **80**. In other words, the spike **85** does not enter the seed aperture **110**.

(22) The point **145** of the spike **85** is configured to not contact the sentinel seed **65** during shaft extension. However, the sidewall **130** may contact the sentinel seed **65** to block the sentinel seed **65**, as described above, in some instances. The spike **85** is configured to not push the sentinel seed **65** into the ground **20**. In some implementations, the spike **85** does not include a channel for transporting the sentinel seed **65** through the spike **85**; rather, the sentinel seed **65** is transported past the spike **85**. Damage to the sentinel seed **65** is inhibited by reducing contact between the spike **85** and the sentinel seed **65**.

(23) As shown in FIG. 10, the seed guide **80** has begun to move back toward the central frame member **40**, tilted toward the first direction **235**, having widened the hole **60**, however the guide first end **95** is still in the ground **20**. The sentinel seed **65**, therefore, has followed the guide channel **120** down into the hole **60**. Misplacement and damage of the sentinel seed **65** is inhibited by injecting the sentinel seed **65** when the guide first end **95** is still in the ground **20**. As shown in FIG. 10, as the seed guide **80** moves toward the central frame member **40**, away from the ground **20**, the spike **85** begins to move back toward the ground **20**, and the planting assembly **70** begins to tilt away from the first direction **235** once more. Because of the tilting of the planting assembly **70**, the

resultant hole **60** is minimally elongated in the first direction **235**. The process repeats to create further holes **60** in series in the first direction **235**. The planting assembly **45** is configured to create one hole **60** on each stroke.

(24) The press wheel **50** (FIG. 2) follows the planting unit and applies a force that redistributes the soil, filling in the holes **60**. The operation leaves a series of loose soil spots, in contrast to the loose trenches created by typical furrow disks. The press wheel **50** may be supported at different distances from the central frame member **40** in order to create different planting heights. The position of the press wheel **50** may be adjusted by actuating the depth control actuator **210** through the associated gage wheels. Extending the depth control actuator **210** lowers the press wheel **50** toward the ground **20**, creating a shallower planting depth, and retracting the depth control actuator **210** raises the press wheel **50** away from the ground **20**, creating a deeper planting depth.

(25) In certain cases, the ground **20** is not uniform. For example, there may be rocks or hard spots embedded in the ground **20**. As the spike **85** punches into the ground **20**, it may run into the hard spot. To inhibit damage to the seeding system **45** and to reduce disruptions to the operation, the force of the rock **R** is deflected through the flexible arm **195**. As illustrated in FIGS. **11-14**, when the spike **85** hits a hard spot, the flexible arm **195** may bend allowing the downward force provided by the driving assembly **75** to continue and the planting assembly **70** to move. In the illustrated implementation the flexible arm **195** bends such that a lower end of the flexible arm **195** moves away from the first direction **235** and the planting assembly **70** is horizontally displaced away from the first direction **235**. However, the flexible arm **195** and the planting assembly **70** may also deflect toward the first direction **235**. The direction of deflection of the flexible arm **195** and the planting assembly **70** may be based on the geometry of the hard spot and the reaction forces within the planting assembly **70**.

(26) The planter design as described above requires less down force than typical furrowing discs. Because the planter **10** has to support less down force, the central frame member **40** can be made of lighter materials. The seeding system **45** design also makes it easy to synchronize the seeding across the rows, making it easy to create a grid planted field. The holes created by the seeding system **45** are relatively small and therefore create less loose soil which makes erosion and subsequent seed loss less likely. Additionally, the driving assembly **75** is driven independently of the motion of the planter **10**. This allows seeding speed **S2** to be varied independently of the planter speed **S1**, allowing for more precise seeding. The small profile of the seeding system **45** and the holes created by the seeding system **45** allows for less distance between row units **30** and therefore less distance between rows. Thus, more rows can be planted.

(27) The planter **10** is operable to simultaneously plant seeds of two or more different plant genotypes in a single planting operation, or in a single pass of the planter **10** through the field **20**. Specifically, the hoppers **35** may each be filled with different seeds of different plant genotypes. Seeds of different plant “genotypes” can include without limitation seeds of different plant varieties, of different plant cultivars, of different plant hybrids, of different plant species, or of different plant characteristics. The different seeds may be distributed to different row units **30** of the planter **10**, or to different seeding systems **45** of a single row unit **30**, or different seeds may be selectively supplied to a single seeding system **45**, as will be further described below. As such, the planter **10** can be utilized to practice intercropping, a multiple cropping practice where two or more crops are grown in proximity. Intercrops may include a first crop, or cash crop, of a first plant type to be harvested, and may further include one or more cover crops of different plant types that are planted between rows of the cash crop or amongst the cash crop. The one or more cover crops can be selected to manage attributes of the soil, such as soil erosion, soil fertility, soil quality, water, weeds, pests, diseases, biodiversity, wildlife, and other attributes.

(28) Additionally, the planter **10** may be used to plant sentinel seeds **65** and other seeds in proximity of, or near, the sentinel seeds **65**. Sentinel seeds **65** exhibit different characteristics when presented with a stressor **240** (FIG. 15). With continued reference to FIG. 15, the different

characteristics can be any detectable plant attribute that is proportional to the stressor **240** such as an abiotic factor **245** or a biotic factor **250**. For example, the abiotic factor **245** may be a lack of or an abundance of water, humidity, light or minerals. The abiotic factor **245** may also be soil conditions or temperature. The biotic factor **250** may be the presence of a certain fungus, bacteria, or insect.

(29) For example, the sentinel seeds **65** may be genetically modified corn or soy beans that exhibit different characteristics from the other corn or soy beans that the sentinel seeds **65** are planted with, respectively. In other examples, the seed may be traditionally bred. The first planting system **12** may be configured for planting a first sentinel plant **255** that exhibits a first sentinel plant characteristic **260**. The first sentinel plant **255** may be a sentinel seed **65** or a sentinel seedling (not shown) or any other sentinel plant form. The first sentinel plant **255** may also be a fungus, a lichen, mold, or any other stationary life form. The second planting system **14** may be configured for planting a second sentinel plant **280** that exhibits a second sentinel plant characteristic **285**. The second sentinel plant characteristic **285** may be the same as the first sentinel plant characteristic **260** or it may vary. The second sentinel plant **280** may be a sentinel seed **65** or a sentinel seedling (not shown) or any other sentinel plant form. The second sentinel plant **280** may also be a fungus, a lichen, mold, or any other stationary life form.

(30) In one embodiment, the first or second sentinel plant characteristic **260**, **285** may include at least one of varying intensity of electromagnetic response **265** generated by the first or second sentinel plant **255**, **280** or a plant growth rate **270** that correspond to a magnitude **275** of the stressor **240**. The electromagnetic response **265** may be an absorption, transmission, backscatter, reflectance, fluorescence, bioluminescence, or other. The electromagnetic response **265** may be induced with or without external stimulation.

(31) In another embodiment, the first or second sentinel plant characteristic **260**, **285** may include a change in a leaf feature **225** such as a change in leaf size, color, color pattern, shape, area index, or temperature. Other leaf features **225** are contemplated by this disclosure.

(32) In yet another embodiment, the first or second sentinel plant characteristic **260**, **285** may include a change in a plant stem feature **370** such as a change in plant stem height, plant stem biomass, plant stem formations or malformations, plant stem diameter, plant stem color, or plant stem color pattern. Other plant stem features **370** are contemplated by this disclosure.

(33) The first or second sentinel plant characteristic **260**, **285** may include the slow plant growth rate **270** or otherwise altered growth rate, or a time to emergence, seed germination rate, or an initiation of reproductive phase.

(34) In another embodiment, the first or second sentinel plant characteristic **260**, **285** may include a change in a flower feature **375** such as flower color or flower timing. Other flower features **375** are contemplated by this disclosure.

(35) In yet another embodiment, the first or second sentinel plant characteristic **260**, **285** may include a senescence feature **380** such as death of the sentinel plant due to daylight, a daylight change derivative, a change in temperature, a soil chemical attribute, or other. Other first or second sentinel plant characteristics **260**, **285** are contemplated by this disclosure.

(36) Referring to FIG. **15**, a block diagram is provided of one example of a computing architecture **290** that includes planter **10**, a sensor **295**, a global positioning system (“GPS”) **300**, and a control system **305**. Global positioning system **300** may comprise a Global Navigation Satellite System (GNSS), a terrestrial radio triangulation system, or any other system which is able to provide the location of the planter **10** in the field **20** in global or local coordinates. FIG. **15** illustratively shows that the planter **10**, the sensor **295**, the GPS **300**, and the control system **305** are connected over a network **310**. Thus, computing architecture **290** operates in a networked environment, where the network **310** includes any of a wide variety of different logical connections such as a local area network (LAN), wide area network (WAN), controller area network (CAN) near field communication network, satellite communication network, cellular networks, or a wide variety of

other networks or combination of networks. It is also noted that the control system **305** can be deployed on the planter **10** such that the control system **305** performs the operations described herein without a networked connection. In addition, while the present description will primarily focus on an example of the control system **305** communicating with the planter **10**, it is noted that the same or similar functionality can be provided when communicating with a wide variety of vehicles **15** and/or remote systems.

(37) The sensor **295** may be communicatively coupled to the planter **10**. For example, the sensor **295** may be coupled to the planter **10** or it may be coupled to a satellite **312** that communicates with the planter **10** or control system **305**. The sensor **295** may be configured for sensing a current field characteristic **315** and generating a current field characteristic signal **320** indicative of the current field characteristic **315** that can be used to determine where to plant the sentinel seeds **65**. The sensor **295** may include an electromagnetic sensor **325** (e.g., optical sensor) and the current field characteristic **315** may include an image **330** that depicts a field condition **335**. The image **330** may be 2D or 3D. The image **330** may comprise a single pixel or data value. The electromagnetic sensor **325** may be a camera or other imaging sensor such as a video camera. Alternatively, the sensor **295** may include at least one of a radar, lidar, laser-based sensors, LIDAR based sensors, temperature sensors, soil property sensors, and a wide variety of other imaging or other sensing systems.

(38) The current field characteristic **315** may include an image **330** that depicts the field condition **335** such as a topography **385** of the field **20** or landscape position **390** of plants **11** or a size of the plant **11** patches. The current field characteristic **315** may also include soil type **395** such as clay composition or top soil composition. The current field characteristic **315** may include a moisture level determination **400** such as 20-80% water content or a nutrient level determination **405** that includes a level of soil organic matter, or residue cover, or cover crops. Additionally, the current field characteristic **315** may include a temperature reading **410** of the field **20** at a soil depth or a pH reading **415** of the soil of the field **20**.

(39) The current field characteristic **315** may include a compaction reading **420** of the field **20** that includes compaction layer details such as depth of the compaction and compaction drainage and its location in the field **20**. Additionally, the current field characteristic **315** may include a weather reading **425**. The weather reading **425** may include past, present, and future temperatures, moisture, solar, sun angle, sunlight attenuation by clouds or other, day length, or other.

(40) The current field characteristic **315** may include a field operations reading **430** that indicates machinery data including machinery settings or timing of machinery operations along with the location in the field **20**. Additionally, the current field characteristic **315** may include a crop reading **435** that indicates crop details including mono or multiple varieties that are planted in the field **20** and their location. The current field characteristic **315** may include a pests reading **440** including an indication of any insects, mammals, birds, or other similar biotic factors and their location in the field **20**. The current field characteristic **315** may include a disease reading **445** including an indication of any bacteria, molds, smuts, viruses or other similar biotic factors and their location in the field. The current field characteristic **315** may include a weed reading **450** including an indication of any weeds and weed seeds and their location in the field **20**. Other current field characteristics **315** are contemplated by this disclosure.

(41) The GPS **300** is communicatively coupled to the planter **10**. The GPS **300** is configured for generating a location signal **340** indicative of a location **345** of the planter **10** or vehicle **15**. Generally, the GPS **300** receives sensor signals from one or more sensors, such as a GPS receiver, a dead reckoning system, a LORAN system, or a wide variety of other systems or sensors, to determine the location **345** of the planter **10** across the field **20**.

(42) The control system **305** is in communication with the sensor **295** and the GPS **300**. The control system **305** is communicatively coupled to the planter **10**. The control system **305** may be configured to receive the first sentinel plant characteristic **260**, receive the second sentinel plant

characteristic **285**, receive the location signal **340**, receive a georeferenced field characteristic **350** from a data storage **355**, and control at least one of the first planting system **12** or the second planting system **14** based on at least one of the first sentinel plant characteristic **260**, the second sentinel plant characteristic **285**, the location **345**, the current field characteristic **315**, or the georeferenced field characteristic **350**, or any combination thereof. The georeferenced field characteristic **350** may include a depth and spacing information for the plants **11** from a prior planting operation or historical data for the field **20**. The georeferenced field characteristic **350** may include relationships between first and second sentinel plant characteristics **260**, **285** that can be sensed and the georeferenced field characteristics **350** that promote a degree of response.

(43) The control system **305** may be configured to control the first planting system **12** to plant the first sentinel plant **255** at the location where the georeferenced field characteristic **350** indicates the stressor **240** for the first sentinel plant characteristic **260**. For example, the georeferenced field characteristic **350** may indicate an area or location of the field **20** where there is or may be a high incidence of an insect. The control system **305** may control the first planting system **12** to plant the first sentinel plant **255** that has the first sentinel plant characteristic **260** that is elicited by the insect at this location. The first sentinel plant characteristic **260** may be the slow growth rate **270**.

(44) In an additional example, the georeferenced field characteristic **350** may indicate an area of the field **20** where there is a high probability of soybean chlorosis in some years. The control system **305** may control the first planting system **12** to plant the first sentinel plant **255** that has the first sentinel plant characteristic **260** that is elicited by difficulties in plant uptake of iron, resulting in chlorosis, at this location. The first sentinel plant characteristic **260** may be a yellowing of leaves proportional to the severity of the stress. Mitigation could be to apply chelated iron and maybe sulfur with the planter **10**, the vehicle **15**, or other equipment.

(45) The georeferenced field characteristic **350** may include a sentinel plant map **360** and the control system **305** may be configured to update the sentinel plant map **360** and the georeferenced field characteristic **350** in the data storage **355** with the current field characteristic **315**. The sensor **295** may be used to obtain the data stored in the data storage **355**. The data storage **355** may be coupled to the planter **10**, the work vehicle **15**, or the satellite **312**. In some examples, satellite **312** may comprise a manned aircraft or unmanned aerial vehicle as a remote sensor platform.

(46) A display **365** may be provided for displaying the sentinel plant map **360** to an operator. Additionally, the operator may enter planter **10** or vehicle **15** commands via the display **365**. Display **365** may include audio elements such as a speaker or microphone. Display may include haptic elements such as a vibration generator.

(47) Referring now to FIG. **16**, a flow diagram of a method **500** for controlling a planter **10** on a field **20** for planting a sentinel plant **11** is provided. At **505**, a first sentinel plant characteristic **260** is received. At **510**, a location signal **340** indicative of a location **345** of the multi-variety planter **10** is received. At **515**, a georeferenced field characteristic **350** is received.

(48) At **520**, a first planting system **12** is controlled to plant the first sentinel plant **255** based on at least one of the first sentinel plant characteristic **260**, the location **345**, or the georeferenced field characteristic **350**. Alternatively, a current field characteristic **315** is sensed and the first planting system **12** is controlled to plant the first sentinel plant **255** based on the first sentinel plant characteristic **260**, the location **345**, and at least one of the georeferenced field characteristic **350** or the current field characteristic **315**. Alternatively, the first planting system **12** may be controlled to plant the first sentinel plant **255** at the location where the georeferenced field characteristic **350** indicates a stressor **240** that may or will cause the first sentinel plant **255** to exhibit the first sentinel plant characteristic **260**. In other examples, the first planting system **12** may be controlled to plant the first sentinel plant **255** at a location which is part of a geographic pattern such as a grid, an elevation gradient, a transect, or other geospatially-dependent scheme.

(49) At **525**, a prescriptive sentinel plant map **535** may be generated. The prescriptive sentinel plant map **535** may include a latitude, a longitude, and a depth of where every plant **11** should be planted

in the field **20**. The prescriptive sentinel plant map **535** may be used by the planter **10** to plant plants **11** including sentinel seeds **65**.

(50) At **530**, the sentinel plant map **360** may be displayed to an operator. The sentinel plant map **360** may be updated to include as-planted data such as where every plant **11** was planted or where and what type of sentinel seeds **65** were planted.

(51) Advantageously, sentinel seeds **65** would not need to be planted at the same time as the main crop. They could be planted at the time of spring tillage ahead of the crop planting pass. They could also be planted after the main crop. In one example, a problem is otherwise identified in the field **20** and then sentinel seeds **65** are planted to play a diagnostic role to find out why the stand is poor or lagging. In another example, sentinel seeds **65** could be planted just prior to a critical crop phase such as pollination to provide data on conditions during a window. Sentinel seeds **65** could also be perennials planted once for multiple seasons of use for annual crops or multi-year crops like cane sugar.

(52) Sentinel seeds **65** could be an organism, other than plants **11**, such as bacteria or fungi (some soil fungi in the tropics are naturally fluorescent). The response of the sentinel organisms could be detected by sensors **295** on a ground engaging element such as a shank. Similarly, the response to a stressor **240** by a sentinel seed **65** may be in the roots and observed below ground.

(53) Sentinel seeds **65** could depend on allelopathy. A first plant, micro, fungi could be the one which is proportionally responsive to the stressor **240**. The response may be in the form of proportionally altering the environment around a second plant. The second plant then communicates the level of that alternate environment. For example, in response to a soil condition, a microbe alters the pH of the soil adjacent to it. This causes a color change in the flower of a co-located plant sensitive to soil pH. A liquid may be precisely applied in the seed trench and planting a sentinel seed **65** may comprise placing a seed and inoculating the soil near the seed with bacteria, fungi, etc. The inoculant wouldn't even need to be alive. It could be the first plant is replaced by a chemical that reacts with one or more soil components, living or non-living, that results in a proportional amount that is reported via the second plant. Chemical may be applied as a liquid, solid, or gas. Also note that the reporter plant response can be other things besides fluorescence such as a nitrogen level experienced by a microbe, a local soil pH level, or a plant flower or leaf color.

(54) Machine learning may be used to improve the efficacy of where to plant plants **11** including sentinel seeds **65**. The ability of the sentinel plants to accurately sense desired crop conditions, in some cases compared to laboratory analysis or other sensing means, may be improved with machine learning. This improvement could be in the selection of varieties to use as sentinels, planting attributes (e.g., interplant spacing, depth), or location of individual plants **11** or plant **11** patches.

Claims

1. A planter configured for planting a sentinel plant in a field, the planter comprising: a first planting system coupled to the planter, the first planting system configured for planting a first sentinel plant having a first sentinel plant characteristic; a global positioning system communicatively coupled to the planter, the global positioning system configured for generating a location signal indicative of a location of the planter; a data storage for storing a georeferenced field characteristic, the georeferenced field characteristic including historical data for the field; a sensor communicatively coupled to the planter, the sensor configured for sensing a current field characteristic and generating a current field characteristic signal indicative of the current field characteristic; and a control system communicatively coupled to the planter, the control system configured to: receive the first sentinel plant characteristic, receive the location signal indicative of the location of the planter, receive the georeferenced field characteristic, receive the current field

characteristic, and control the first planting system based on the first sentinel plant characteristic, the location, and at least one of the georeferenced field characteristic and the current field characteristic to plant the first sentinel plant at the location that includes a stressor that elicits the first sentinel plant characteristic.

2. The planter of claim 1, further comprising a second planting system coupled to the planter, the second planting system configured for planting a second sentinel plant having a second sentinel plant characteristic, the control system configured to receive the second sentinel plant characteristic, control at least one of the first planting system or the second planting system based on at least one of the first sentinel plant characteristic, the second sentinel plant characteristic, the location, and at least one of the georeferenced field characteristic and the current field characteristic to plant the first sentinel plant or the second sentinel plant at the location that includes a stressor that elicits the first sentinel plant characteristic or the second sentinel plant characteristic.

3. The planter of claim 1, wherein the georeferenced field characteristic comprises a sentinel plant map and the control system is configured to update the sentinel plant map with the current field characteristic.

4. The planter of claim 3, further comprising a display for displaying the sentinel plant map to an operator.

5. The planter of claim 1, wherein the first sentinel plant characteristic comprises at least one of varying intensities of electromagnetic response generated by a sentinel plant or a plant growth rate that correspond to a pressure of the stressor.

6. A multi-variety planter configured for planting a sentinel plant in a field, the multi-variety planter comprising: a first planting system coupled to the multi-variety planter, the first planting system configured for planting a first sentinel plant having a first sentinel plant characteristic; a second planting system coupled to the multi-variety planter, the second planting system configured for planting a second sentinel plant having a second sentinel plant characteristic; a global positioning system communicatively coupled to the multi-variety planter, the global positioning system configured for generating a location signal indicative of a location of the multi-variety planter; a data storage for storing a georeferenced field characteristic, the georeferenced field characteristic includes historical data for the field; a sensor communicatively coupled to the planter, the sensor configured for sensing a current field characteristic and generating a current field characteristic signal indicative of the current field characteristic; and a control system communicatively coupled to the multi-variety planter, the control system configured to: receive the first sentinel plant characteristic, receive the second sentinel plant characteristic, receive the location signal indicative of the location of the planter, receive the georeferenced field characteristic, receive the current field characteristic; and control at least one of the first planting system or the second planting system based on the first sentinel plant characteristic, the second sentinel plant characteristic, the location, and at least one of the georeferenced field characteristic and the current field characteristic to plant the first sentinel plant or the second sentinel plant at the location that includes a stressor that elicits the first sentinel plant characteristic or the second sentinel plant characteristic.

7. The multi-variety planter of claim 6, wherein the sensor comprises an electromagnetic sensor, the current field characteristic comprises an image that depicts a field condition.

8. The multi-variety planter of claim 6, wherein the georeferenced field characteristic comprises a sentinel plant map and the control system is configured to update the sentinel plant map with the current field characteristic.

9. The multi-variety planter of claim 8, further comprising a display for displaying the sentinel plant map to an operator.

10. The multi-variety planter of claim 6, wherein the first sentinel plant characteristic comprises at least one of varying intensities of electromagnetic response generated by a sentinel plant or a plant

growth rate that correspond to a pressure of the stressor.

11. The multi-variety planter of claim 10, wherein the stressor comprises at least one of an abiotic factor or a biotic factor.

12. A method of controlling a planter for planting a sentinel plant in a field, the method comprising: receiving a first sentinel plant characteristic; receiving a location signal indicative of a location of the planter; receiving a georeferenced field characteristic from a data storage, the georeferenced field characteristic including historical data for the field; receiving a current field characteristic; and controlling a first planting system to plant the first sentinel plant based on the first sentinel plant characteristic, the location, and at least one of the georeferenced field characteristic and the current field characteristic to plant the sentinel plant at the location that includes a stressor that elicits the first sentinel plant characteristic.

13. The method of claim 12, wherein the first sentinel plant characteristic comprises at least one of varying intensities of electromagnetic response generated by a sentinel plant or a plant growth rate that correspond to a pressure of the stressor.

14. The method of claim 12, further comprising displaying a sentinel plant map to an operator.
